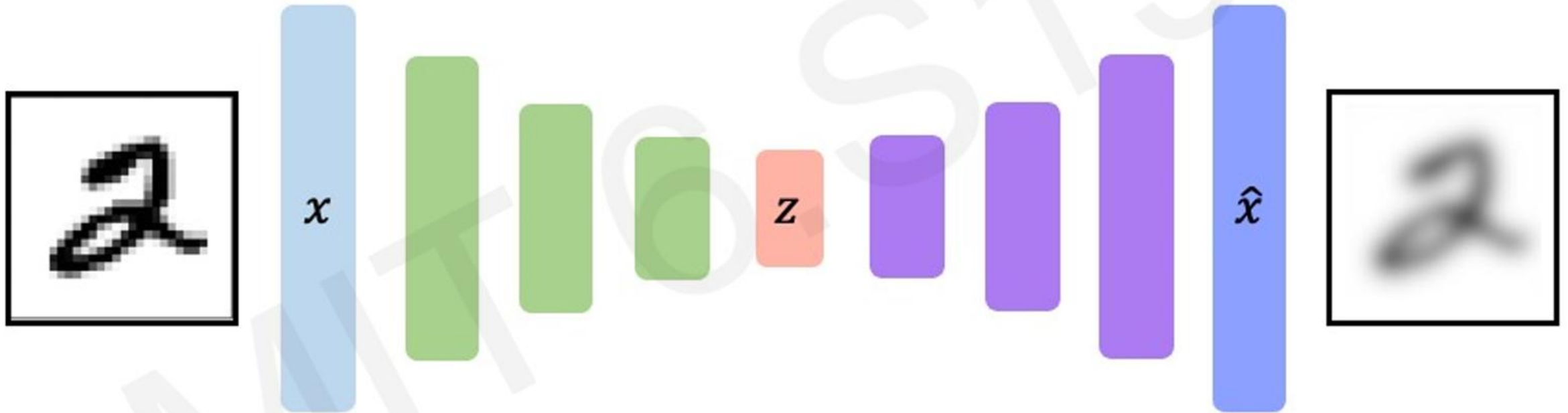


Autoencoders: background

How can we learn this latent space?

Train the model to use these features to **reconstruct the original data**



“Decoder” learns mapping back from latent space, z , to a reconstructed observation, $\hat{x}$

What is a latent variable?

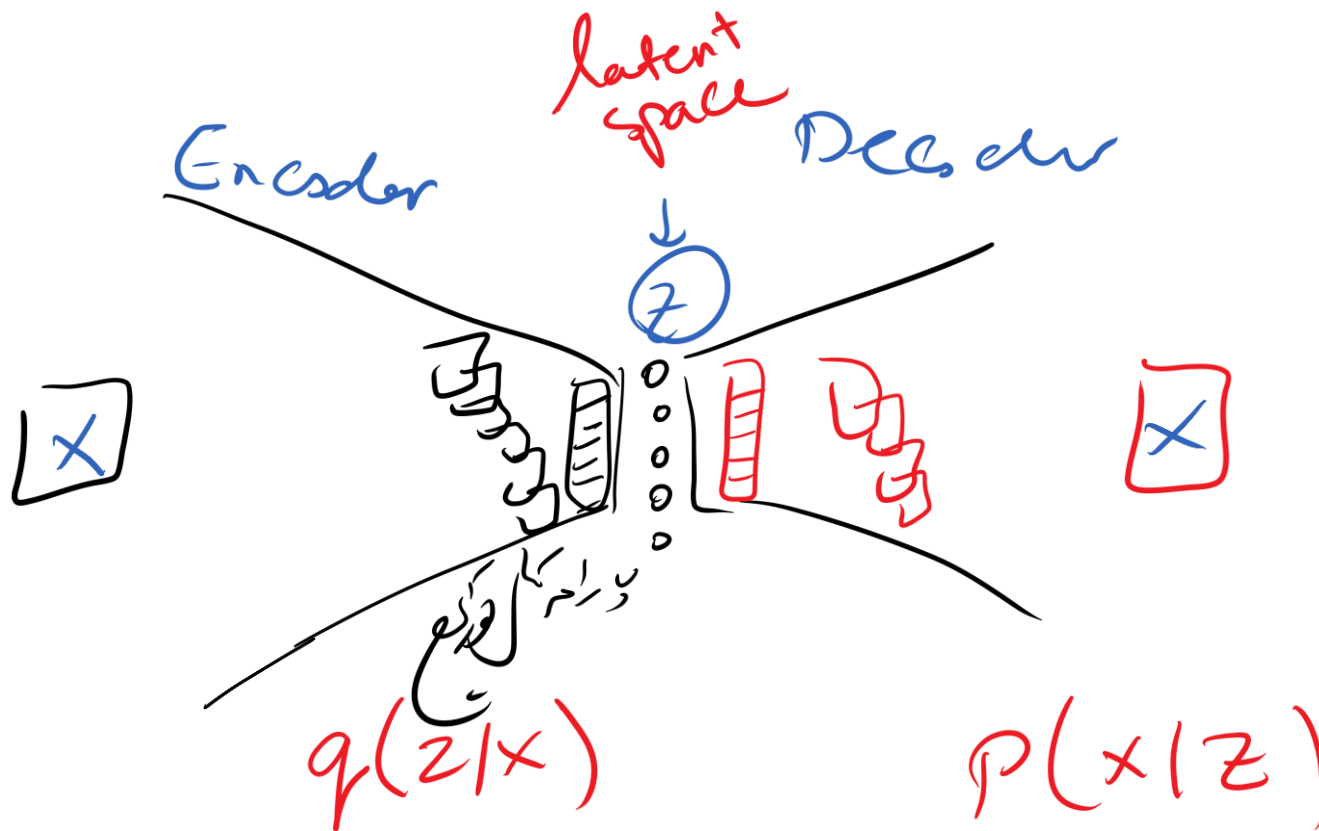


Can we learn the **true explanatory factors**, e.g. latent variables, from only observed data?

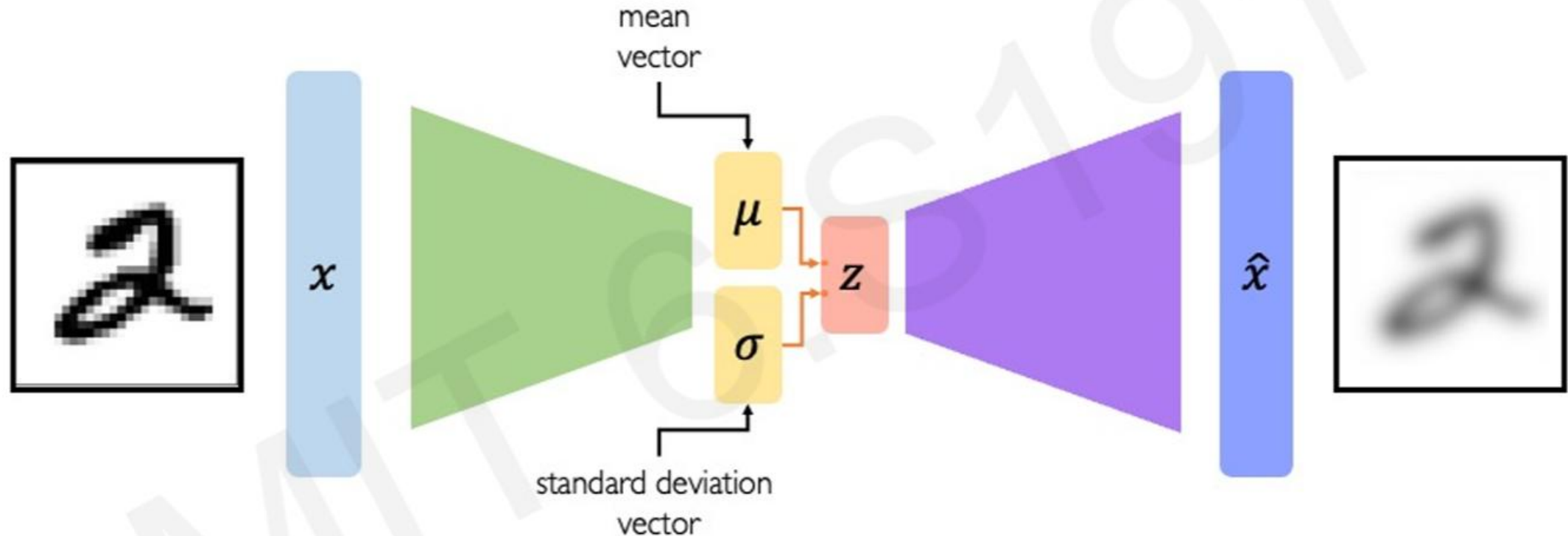


KL

Variational
Auto Encoder



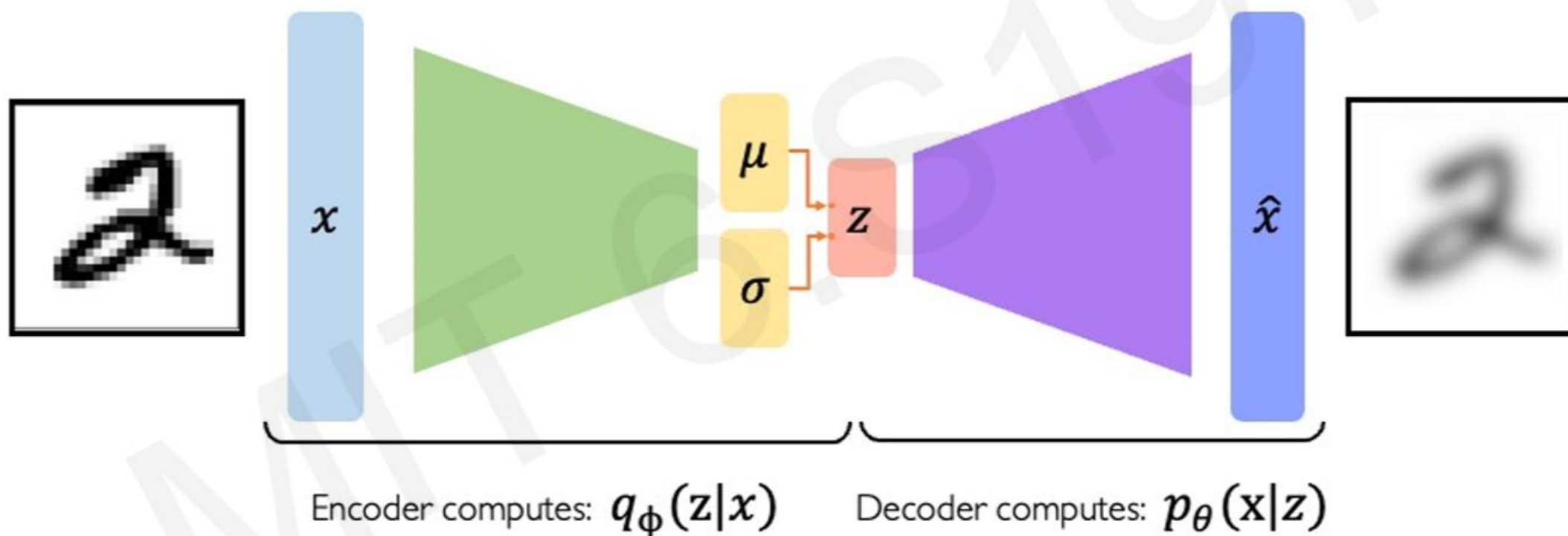
VAEs: key difference with traditional autoencoder



Variational autoencoders are a probabilistic twist on autoencoders!

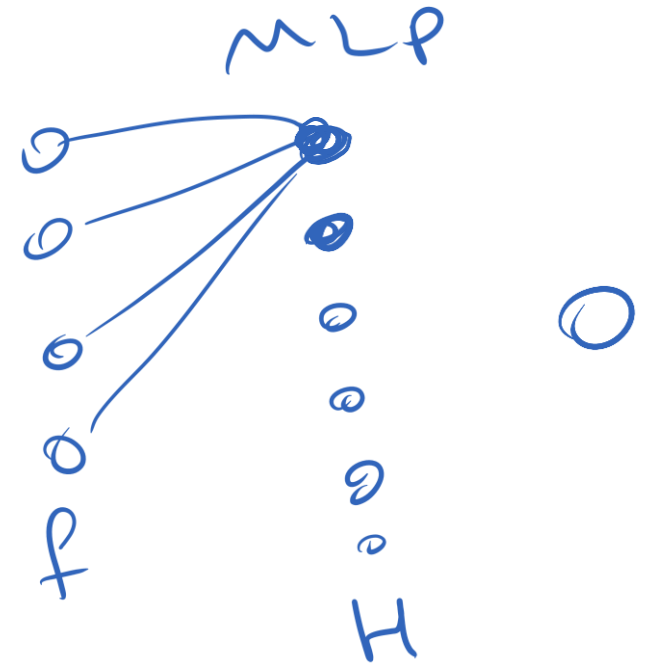
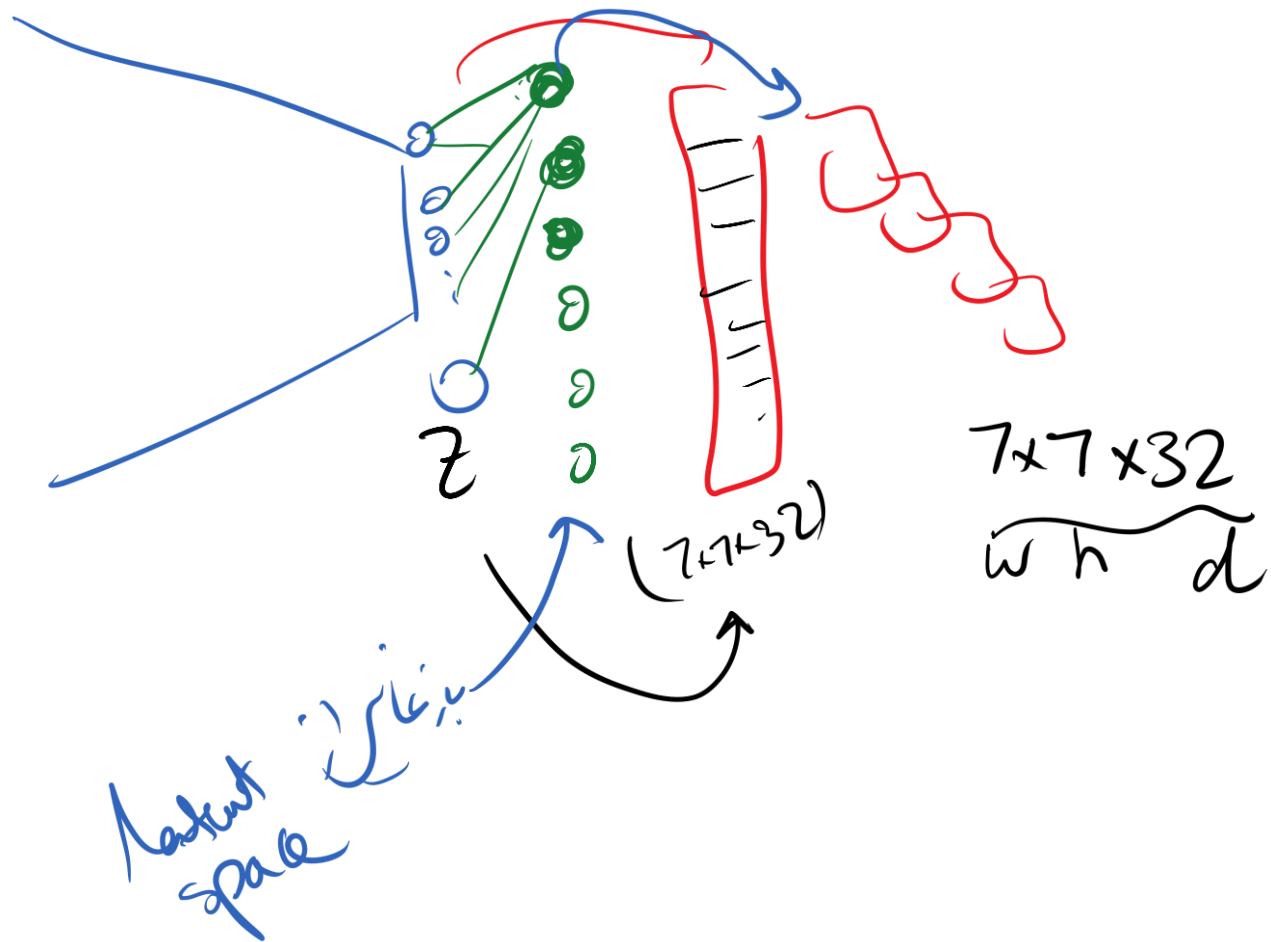
Sample from the mean and standard deviation to compute latent sample

VAE optimization



$$\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & \phi & \sigma_3 & \\ & & \dots & \sigma_f \\ & & & & \sigma_f \end{bmatrix}$$

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_f \end{bmatrix}$$



Standard Scaler

$$\mathcal{N}(\mu, \sigma) \xrightarrow{\frac{x - \mu}{\sigma}} \mathcal{N}(0, 1)$$

$$\mathcal{N}(\mu, \sigma) \quad \overleftarrow{Z = \mu + \sigma \times \varepsilon}$$

$\varepsilon \sim \mathcal{N}(0, 1)$

GAN

Random



Decoder
Generator



Test

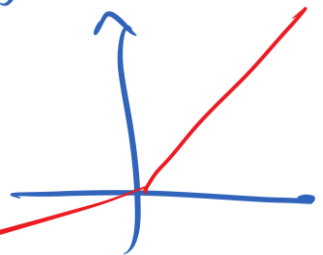
CNN Classifier

Discriminator

Real

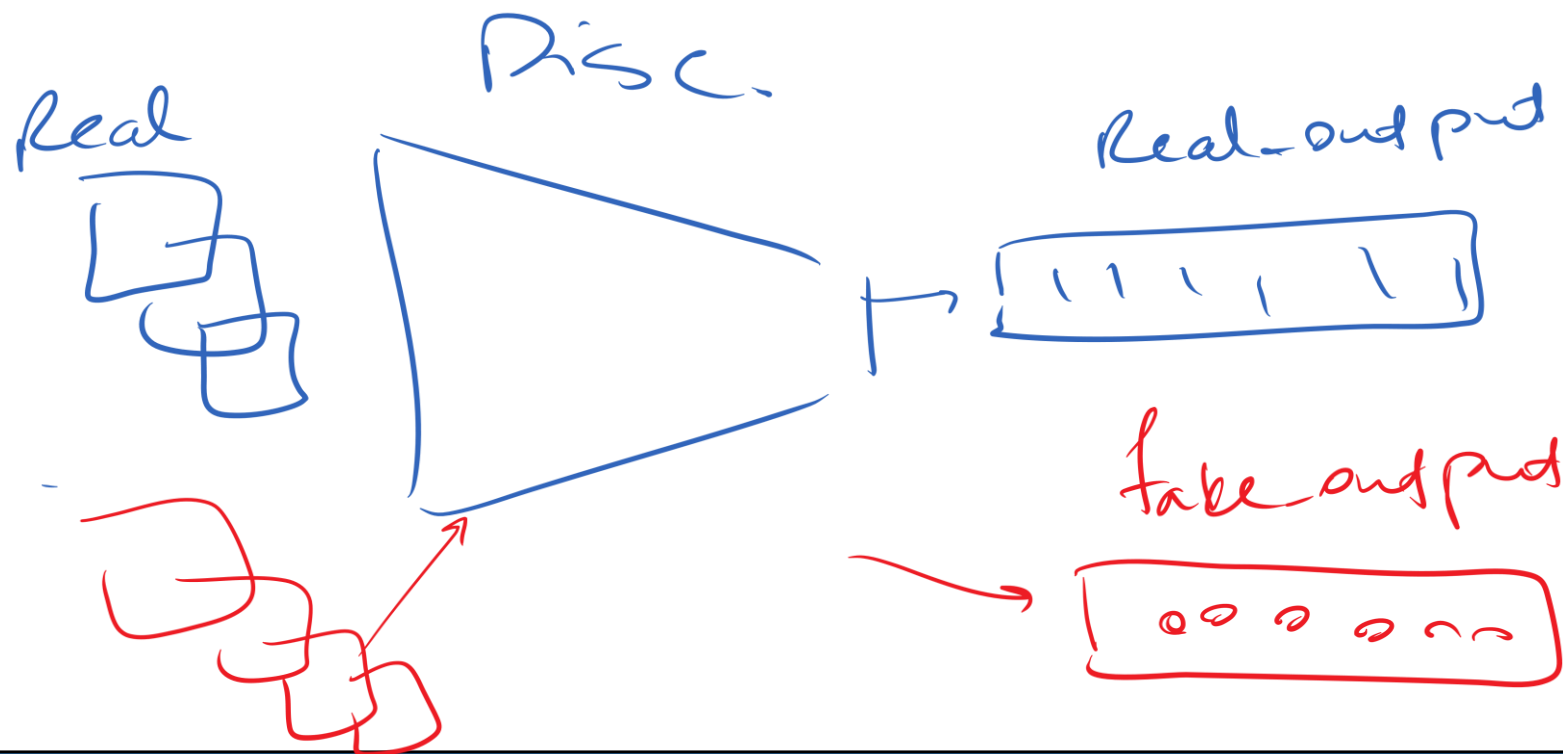
Fake

∞

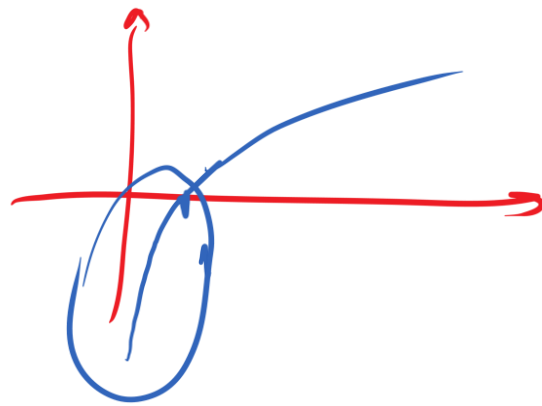


Gen

$$w_N = w_0 - \eta \frac{\partial L}{\partial w}$$



Information: $-\log p(x)$
Curl



Entropy: $-\sum p(x) \log p(x)$

$KL_{(p,q)}$: $-\sum \frac{p(x)}{q(x)} \log(q(x)) + \sum p(x) \log p(x)$

$KL(\hat{p} || q) = -\sum p(x) \log \frac{q(x)}{p(x)}$

$KL \geq 0$



$$P(z|x) = \frac{P(x, z)}{P(x)}$$

توزیع مشروط بر x

$$\rightarrow q(z)$$

توزیع طم اعتبار

$$\text{Min KL}(q(z) || P(z|x)) = - \sum q(z) \log \frac{P(z|x)}{q(z)}$$

$$= - \sum q(z) \log \frac{P(x, z)}{P(x) q(z)} = - \sum q(z) \log \left(\frac{P(x, z)}{q(z)} \times \frac{1}{P(x)} \right)$$

$$- \sum q(z) \log \left(\frac{P(x, z)}{q(z)} \times \frac{1}{P(x)} \right) = - \sum q(z) \left[\underbrace{\log \frac{P(x, z)}{q(z)}}_{\text{KL}} - \underbrace{\log P(x)}_{\text{prior}} \right]$$

$$= - \sum_z q(z) \log \frac{P(x, z)}{q(z)} + \underbrace{\sum_z q(z) \log (P(x))}_{\downarrow}$$

$$- \sum q(z) \log \frac{P(x, z)}{q(z)} + \log P(x) \underbrace{\sum_z q(z)}_{1}$$

$$KL(q(z) || P(z|x)) = - \sum q(z) \log \frac{P(x, z)}{q(z)} + \log P(x)$$

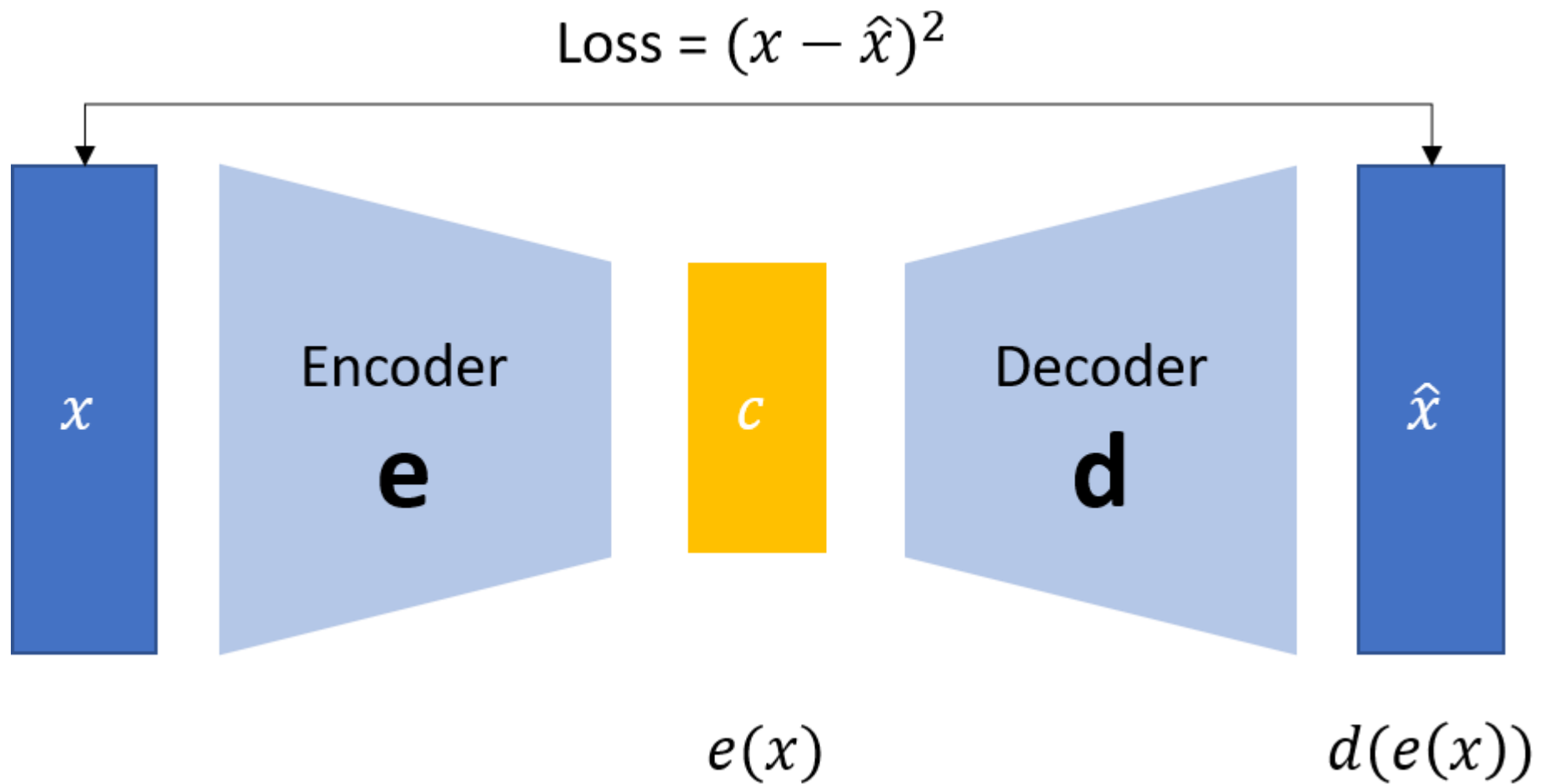
$$KL(q(z) || p(z|x)) = - \sum q(z) \log \frac{p(x,z)}{q(z)} + \log p(x)$$

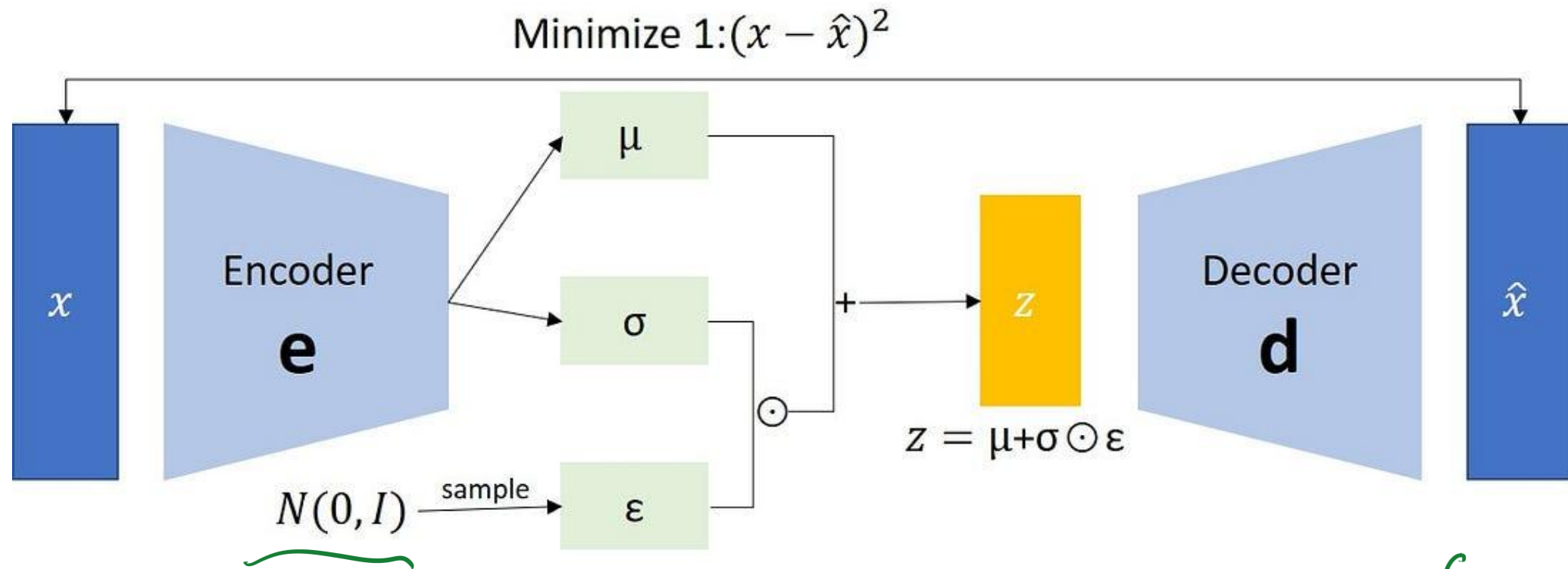
$$\underbrace{KL(q(z) || p(z|x))}_{\downarrow} + \underbrace{\sum_z q(z) \log \frac{p(x,z)}{q(z)}}_{\substack{\uparrow \\ \text{ELBO} \\ L}} = \log(p(x))_{\text{fixed}}$$

$$L = \sum q(z) \log \frac{p(x|z)}{q(z)} = \sum q(z) \log \frac{p(x|z)p(z)}{q(z)}$$

$$= \sum q(z) \left(\log p(x|z) + \log \frac{p(z)}{q(z)} \right)$$

$$= \underbrace{\sum q(z) \log p(x|z)}_{\substack{E_{q(z)}[\log p(x|z)]}} + \underbrace{\sum q(z) \log \frac{p(z)}{q(z)}}_{-KL(q(z) \parallel p(z))}$$





Minimize 2: $\frac{1}{2} \sum_{i=1}^N (\exp(\sigma_i) - (1 + \sigma_i) + \mu_i^2)$

$$z = \mu + \sigma \odot \varepsilon$$

$$\varepsilon = \frac{n - \mu}{\sigma}$$

$$P(x) = \int_z P(z) P(x|z) dz$$

Maximum $L = \sum_x \log P(x)$

$$\begin{aligned}
\log P(x) &= \int_z q(z|x) \log P(x) dz \\
&= \int_z q(z|x) \log \left(\frac{P(z, x)}{P(z|x)} \right) dz \\
&= \int_z q(z|x) \log \left(\frac{P(z, x)}{q(z|x)} \frac{q(z|x)}{P(z|x)} \right) dz \\
&= \int_z q(z|x) \log \left(\frac{P(z, x)}{q(z|x)} \right) dz + \int_z q(z|x) \log \left(\frac{q(z|x)}{P(z|x)} \right) dz \\
&= \int_z q(z|x) \log \left(\frac{P(z, x)}{q(z|x)} \right) dz + KL(q(z|x) || P(z|x))
\end{aligned}$$

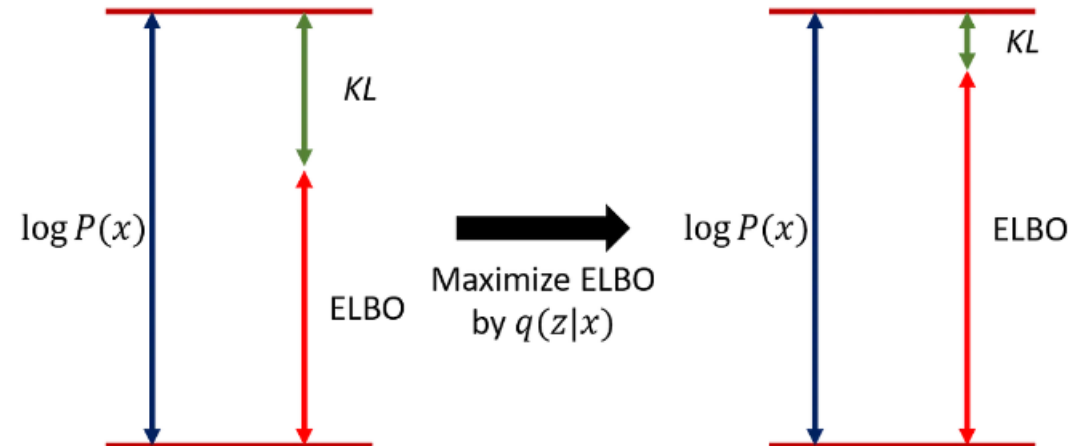
$$\log P(x) \geq \int_z q(z|x) \log \left(\frac{P(x|z)P(z)}{q(z|x)} \right) dz$$

We denote this lower bound as ELBO:

$$\log P(x) \geq \text{ELBO} = \int_z q(z|x) \log \left(\frac{P(x|z)P(z)}{q(z|x)} \right) dz = \mathbb{E}_{q(z|x)} \left[\log \frac{p(x, z)}{q(z|x)} \right]$$

So we can revise the original form as:

$$\log P(x) = \text{ELBO} + KL(q(z|x) || P(z|x))$$



$$\begin{aligned}
\text{ELBO} &= \int_z q(z|x) \log \left(\frac{P(z, x)}{q(z|x)} \right) dz \\
&= \int_z q(z|x) \log \left(\frac{P(x|z)P(z)}{q(z|x)} \right) dz \\
&= \int_z q(z|x) \log \left(\frac{P(z)}{q(z|x)} \right) dz + \int_z q(z|x) \log P(x|z) dz \\
&= -KL(q(z|x) || P(z)) + \int_z q(z|x) \log P(x|z) dz
\end{aligned}$$

$$\text{Maximum} \int_z q(z|x) \log P(x|z) dz$$

$$\text{Maximum } E_{q(z|x)} [\log P(x|z)]$$

$$N(\mu, \sigma)$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right)$$

$$\int_z q(z|x) \log P(z) dz = \int_z N(z; \mu, \sigma^2) \log N(z; 0, I) dz$$

$$= -\frac{J}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^J (\mu_i^2 + \sigma_i^2)$$

$$\int_z q(z|x) \log q(z|x) dz = \int_z N(z; \mu, \sigma^2) \log N(z; \mu, \sigma^2) dz$$

$$= -\frac{J}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^J (1 + \log \sigma_i^2)$$

Then, we can write $-KL(q(z|x)||P(z))$ as:

$$-KL(q(z|x)||P(z)) = \int_z q(z|x) (\log P(z) - \log q(z|x)) dz$$

$$= -\frac{1}{2} \sum_{i=1}^J (1 + \log(\sigma_i^2) - \mu_i^2 - \sigma_i^2)$$

$$N(\mu, \Sigma)$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ & 0 & 0 & 0 & 0 \\ & & 0 & 0 & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{bmatrix}$$